FIITJEE **JEE (Main)-2018**

ALL INDIA TEST SERIES

FULL TEST – VIII

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

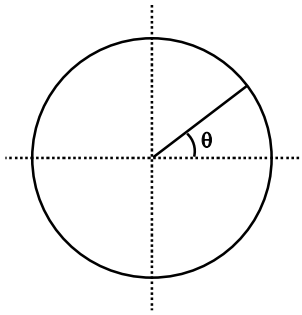
SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

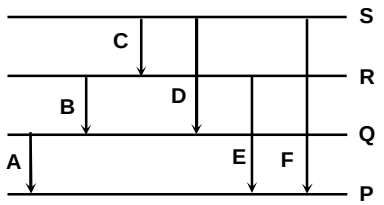
1. A nucleus is kept at rest. It breaks into two parts which fly off with velocity in the ratio 27 : 8. Then the ratio of radii of the parts is
   (A) 27 : 8 (B) 8 : 27
   (C) 2 : 3 (D) 3 : 2

2. A monochromatic light of wavelength 6500 Å is used in YDSE. Now both slits are covered by two thin slits of refractive index 3.2 and 2.7 respectively. By doing so central bright fringe shifts to original 7th bright fringe. If both slits have same thickness then it is equal to
   (A) 7.8 μm (B) 9.1 μm
   (C) 6.5 μm (D) 9.8 μm

3. A non conducting ring of radius 4 cm has varying linear charge density which is given by $\lambda = 3 \cos \theta$ C/m. The net electric dipole moment of ring is give by
   (A) 48 π Coulomb-meter
   (B) 12 $\pi \times 10^{-4}$ Coulomb-meter
   (C) 12 π Coulomb-meter
   (D) 48 $\pi \times 10^{-4}$ Coulomb-meter



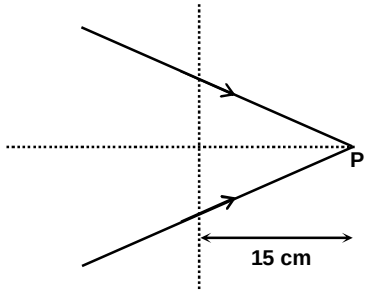
4. Figure shows six spectral lines in a emission spectrum level 'P' is a ground state and Q, R, S are upper excited state. The spectral line also occur in absorption spectrum are
   (A) A, B, C, D, E, F
   (B) A, B, C
   (C) A, E, F
   (D) B, C, D



Space for Rough work

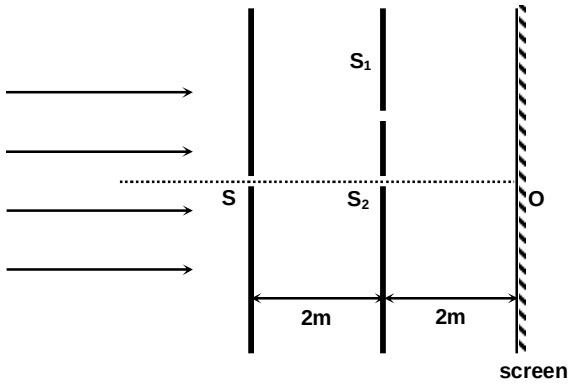
5. E, L, p and r are the energy, angular momentum, linear momentum and orbital radius of an electron in a atom corresponding to the quantum number n. Then EprL is
(A) proportional to n^2 (B) proportional to n.
(C) proportional to 1/n (D) independent of n.

6. A beam of light is converging at a point 'P'. A plano concave lens of refractive index 4/3 and radius of curvature 10 cm is placed in the path of beam at O such that line OP is the principal axis of lens. Then



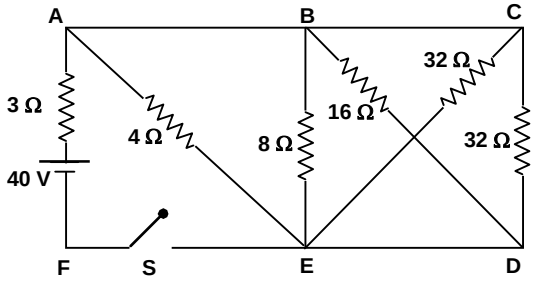
(A) Light rays will diverge
(B) Light rays will converge before point 'P' at a distance 10 cm from O.
(C) Light rays will converge after point P at 30 cm from the point P.
(D) Light rays will converge after point P at 15 cm from the point P.

7. A parallel beam of light of wavelength 5000 Å is incident on slit S. Another two slits S_1 and S_2 are illuminated at a distance 2 meter as shown in the figure. The minimum distance between S_1 and S_2 such that maximum intensity will occur at O is



(A) $\sqrt{10}$ mm
(B) 1 mm
(C) $\frac{1}{2}$ mm
(D) $\sqrt{5}$ mm

8. Six resistors and a battery are connected as shown in the figure. Now switch S is closed then find the ratio of current in branch CD to current in branch AF.



(A) 16
(B) 8
(C) $\frac{1}{16}$
(D) $\frac{1}{8}$

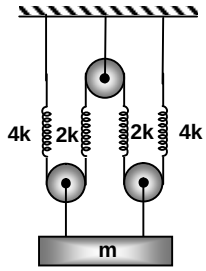
Space for Rough work

9. The field normal to the plane of coil of N turns, radius r carrying current i at a point lying a small distance h (<< r) from the centre of coil is B_1 and field at the centre of coil is B_2. Then the value of $\frac{B_2 - B_1}{B_2}$ is

(A) $\frac{2h^2}{3r^2}$ (B) $\frac{2h^2}{r^2}$

(C) $\frac{h^2}{3r^2}$ (D) $\frac{3h^2}{2r^2}$

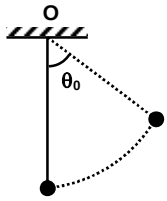
10. A plank of mass M is connected with three massless pulleys and four massless springs as shown in the figure in vertical plane. Plank is slightly displaced from its mean position. Assuming plank remain in horizontal orientation throughout the motion, the frequency of oscillation of plank is



(A) $\frac{1}{2\pi}\sqrt{\frac{32K}{3m}}$ (B) $\frac{1}{2\pi}\sqrt{\frac{8K}{3m}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{4K}{3m}}$ (D) $\frac{1}{2\pi}\sqrt{\frac{16K}{3m}}$

11. A small bob hangs from a support 'O' in a vertical plane if the bob is slightly displaced then it oscillate about its mean position as shown in figure. The angular amplitude of the pendulum is θ_0. If the ratio of maximum tension to minimum tension during the motion is 4 than angular amplitude is given by

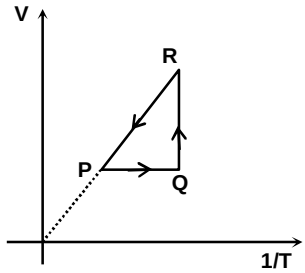


(A) $\theta_0 = \cos^{-1}\left(\frac{1}{4}\right)$ (B) $\theta_0 = \cos^{-1}\left(\frac{3}{4}\right)$

(C) $\theta_0 = \cos^{-1}\left(\frac{1}{2}\right)$ (D) $\theta_0 = \cos^{-1}\left(\frac{3}{5}\right)$

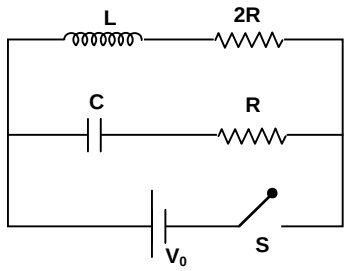
12. An ideal gas undergoes a cyclic process as shown in figure. Regarding the process choose the correct statement.

(A) Heat energy is rejected in process PQ and QR.
(B) Pressure of gas decreases in process RP.
(C) Heat energy must be rejected in process RP.
(D) Heat energy may be absorbed in process RP.



Space for Rough work

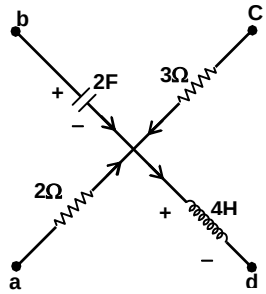
13. An inductor of inductance L and two resistors of resistance R and 2R are connected in a circuit as shown in figure. Now, switch 'S' is closed at t = 0. The time at which potential drop across inductor is half of the potential drop across R is

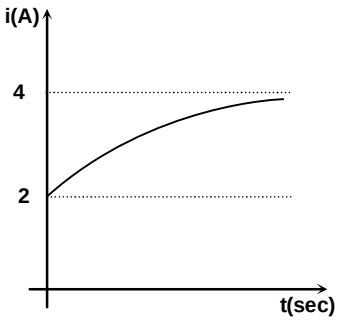


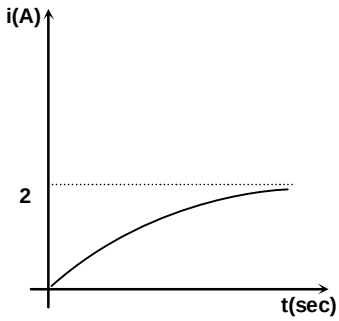
(A) $\frac{LRC\ell n2}{2R^2C+L}$ (B) $\frac{LCR\ell n2}{2R^2C-L}$

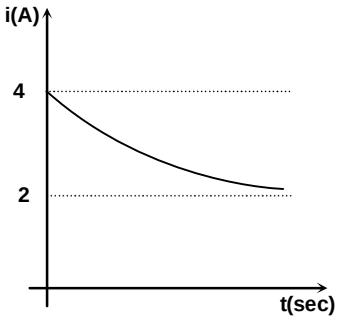
(C) $\frac{LCR\ell n2}{R^2C+2L}$ (D) $\frac{LCR\ell n2}{R^2C-2L}$

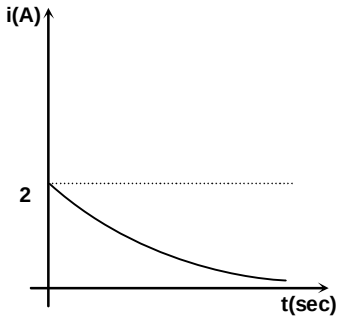
14. Figure shown is a part of a given circuit. The current in 3 Ω resistor and 2 Ω resistor is given by 4 amp and $10e^{-2t}$ amp respectively. The voltage across the capacitor is given by $3e^{-2t}$ volt. The graph between current versus time in the inductor is given by



(A)


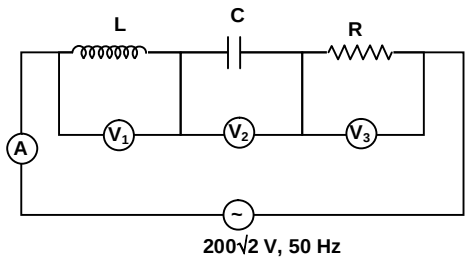
(B)


(C)


(D)


Space for Rough work

15. A inductor of inductive reactance 100 Ω, a capacitor of capacitive reactance 50 Ω and resistor of resistance R Ω are connected in a series as shown in figure. All voltmeters and ammeter connected in circuit are ideal. The reading of voltmeter V_1 and V_2 is 400 Volt and 200 Volt respectively and reading of ammeter is 4 amp. The reading of voltmeter V_3 and resistance of resistor are



(A) 100 V, 100 Ω (B) $100\left(2\sqrt{2}-1\right)$ V, 50Ω

(C) $100(2\sqrt{2}-1), 100\Omega$ (D) 200 V, 50 Ω

16. A plane longitudinal wave having angular frequency ω = 500 rad/sec is travelling in negative x-direction in a medium of density ρ = 1 kg/m^3 and bulk modulus 4×10^4 N/m^2. The loudness at a point in medium is observed to be 20 dB. Assuming at x = 0 initial phase of the medium particles to be zero. The maximum pressure charge in the medium and equation of wave is given by

(A) 2×10^{-4} N/m^2, $y = 2 \times 10^{-9} \sin\left(500t + \frac{5}{2}x\right)$ (B) 10^{-4} N/m^2, $y = 2 \times 10^{-9} \sin\left(500t + \frac{5}{2}x\right)$

(C) 4×10^{-4} N/m^2, $y = 4 \times 10^{-9} \sin(500\,t + 5x)$ (D) 2×10^{-4} N/m^2, $y = 4 \times 10^{-9} \sin\left(500t + \frac{5}{2}x\right)$

17. Two resistors of resistance 400 Ω and 800 Ω at 0°C, having thermal coefficient of resistance α and 4 α respectively are connected in parallel. The equivalent thermal coefficient of resistance is

(A) 5 α (B) $\frac{3\alpha}{2}$

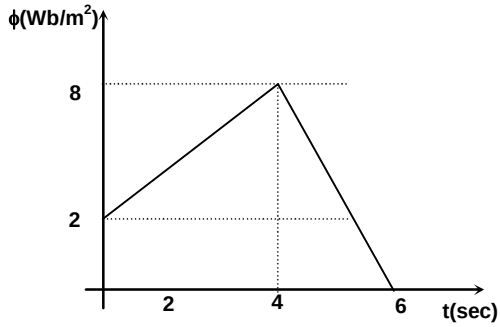
(C) 2 α (D) $\frac{4\alpha}{5}$

18. If $\oint \vec{E} \cdot d\vec{s} = 0$ over a surface then

(A) the electric field inside the surface and on it must be zero.
(B) the electric field inside the surface is necessarily uniform.
(C) all charges may be inside the surface.
(D) The number of flux lines entering the surface must be equal to the number of flux lines leaving it.

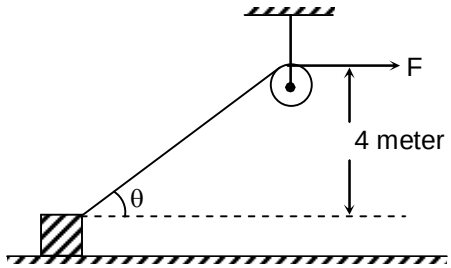
Space for Rough work

19. The magnetic flux (ϕ) versus time (t) graph for a coil is shown in the figure. The magnitude of average emf induced in the coil for a time interval of 0 to 6 sec is
(A) 1 Volt
(B) 1/3 Volt
(C) 2/3 Volt
(D) 2 Volt



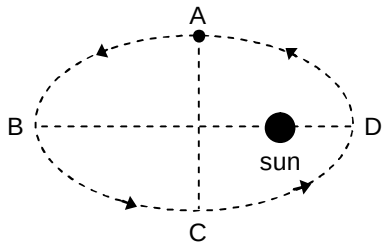
20. If the gravitational constant 'G'. Plank's constant 'h' and velocity of light 'v' are assumed to be as fundamental units then dimension of length in new system is given by
(A) G^2, h^2, $v^{3/2}$ (B) $G^{1/2}$, $h^{1/2}$, $v^{3/2}$
(C) $G^{1/2}$, $h^{1/2}$, $v^{-3/2}$ (D) G^2, h^2, $v^{-3/2}$

21. A block of mass 2 kg is kept at rest on a rough horizontal surface as shown in the figure. A constant force of magnitude 20N is applied at other end of string in horizontal direction. Initially angle made by string with horizontal is 30°. The work done by applied force till the angle become 45° is



(A) $80\left(2-\sqrt{2}\right)$ Joule (B) $80\sqrt{2}$ Joule
(C) $40\left(2-\sqrt{2}\right)$ Joule (D) $40\sqrt{2}$ Joule

22. A satellite is moving around the sun in an elliptical orbit. 'M' is the mass of sun and 'a' is semi major axis of elliptical orbit. The velocity of satellite when it reaches at A is
(A) $\sqrt{\frac{4GM}{a}}$ (B) $\sqrt{\frac{2GM}{a}}$
(C) $\sqrt{\frac{GM}{2a}}$ (D) $\sqrt{\frac{GM}{a}}$

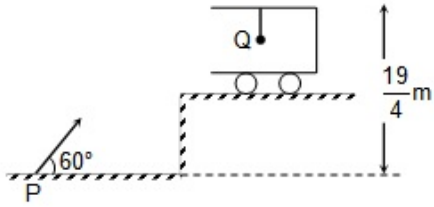


Space for Rough work

23. A particle is projected from the ground at an angle with the horizontal. The velocity of the projectile when it is at the greatest height is $\sqrt{\frac{3}{5}}$ times when it is at one third of its greatest height. The angle of projection should be

(A) 37° (B) 45°
(C) 53° (D) 60°

24. A small ball 'P' of mass 2 kg is projected with velocity 10 m/s at an angle 60° with horizontal. It collides inelastically with another ball 'Q' of same mass. Ball 'Q' is suspended through an inelastic string of length 1m with the ceiling of a trolley of mass 16 kg as shown in figure. Initially trolley is at rest and it can move without friction on the surface. The maximum angle made by thread with vertical is



(A) $\cos^{-1}(2/5)$ (B) $\cos^{-1}(2/3)$
(C) $\cos^{-1}(3/4)$ (D) $\cos^{-1}(3/5)$

25. In forced oscillations, sometimes it is important to analyze the effect of periodically varying force (say, $F = F_0 \cos \omega t$). So the equation becomes

$(F_0 \cos \omega t) + (-kx) = Ma$

A steady state solution of the above equation takes the form $x(t) = A \cos \omega t$. Let δ be a real number such that $\delta \to 0^+$. Then

(A) If $\omega = \delta$, force and displacement will be in phase.

(B) If $\omega = \sqrt{\frac{k}{m}} - \delta$, force and displacement will be out of phase.

(C) If $\omega = \sqrt{\frac{k}{m}} + \delta$, force and displacement will be in phase.

(D) If $\omega \to \infty$, force and displacement will be out of phase.

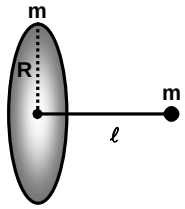
Space for Rough work

26. A uniform disc (mass M, radius R, thickness t) is kept on a rough surface with angular speed ω (without any translation) such that its plane is horizontal. Due to heat generated by friction, the disc gets heated up from an initial temperature T. Since different parts of the disc have different energies, the disc gets heated up non-uniformly (assuming no heat exchange between concentric layers). Once the disc stops, a point on the perimeter has temperature of $T + \Delta T$. If there are no losses to the environment, then what is the temperature at a point exactly midway between centre and perimeter?

(A) $T + \frac{\Delta T}{4}$ (B) $T + \frac{\Delta T}{8}$

(C) $T + \frac{\Delta T}{2}$ (D) $T + \frac{\Delta T}{16}$

27. A thin homogeneous disc of radius R and mass m and a small sphere of same mass is placed at a distance ℓ from the disc on its axis of symmetry. Initially both are in rest and then released from rest. They collide because of gravitation attraction. Assume $\ell >> R$, Find the magnitude of relative velocity of approach just before the collision.



(A) $\sqrt{4Gm\left(\frac{2}{R}-\frac{1}{\ell}\right)}$ (B) $\sqrt{2Gm\left(\frac{2}{R}-\frac{1}{\ell}\right)}$

(C) $\sqrt{Gm\left(\frac{2}{R}-\frac{1}{\ell}\right)}$ (D) $\sqrt{\frac{Gm}{2}\left(\frac{2}{R}-\frac{1}{\ell}\right)}$

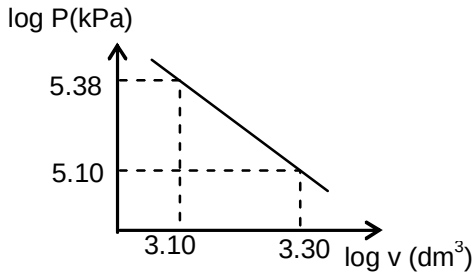
28. Logarithms of reading of pressure and volume for an ideal gas were plotted in the graph. By measuring the gradient it can be shown that the gas may be

(A) Mono atomic and undergoing an adiabatic change

(B) Mono atomic and undergoing an isothermal change

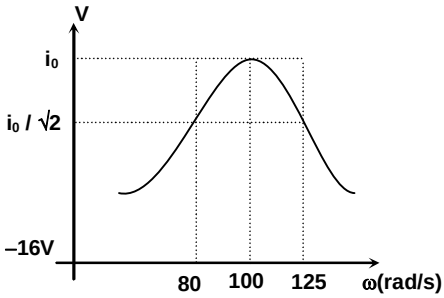
(C) Diatomic and undergoing an adiabatic change

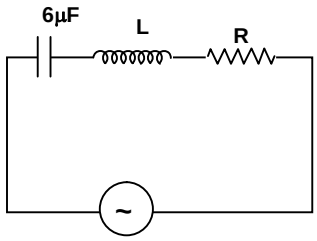
(D) Triatomic and undergoing an isothermal change.



Space for Rough work

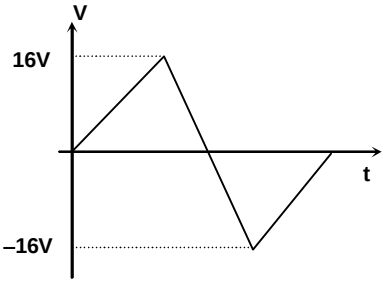
29. An LCR series circuit is connected to an sinusoidal AC voltage and the graph between current and angular frequency is plotted as shown in the figure. If capacitance of capacitor used is 6 μF, then choose the correct alternative.

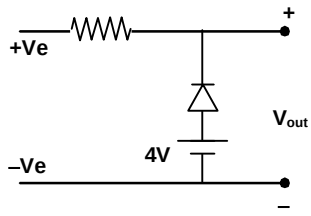




(A) L = 16.6 H, R = 1.5 kΩ
(B) L = 3.34 H, R = 1.5 kΩ
(C) L = 16.67 H, R = 0.75 kΩ
(D) can't be calculated

30. A circuit is designed using ideal diode and input voltage is fed as shown in figure. Maximum and minimum value across output terminal will be





(A) +16V, –16V
(B) +16V, –4V
(C) +12V, –4V
(D) +16V, +4V

Space for Rough work

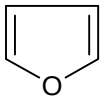
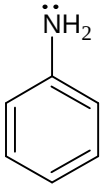
Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following is correctly matched?

	Molecule	Direction of dipole moment
(A)		
(B)		
(C)	H—C≡C—Cl	
(D)	None of these	

32. Electronic configuration of Pd is $[Kr]4d^{10}$. It belongs to
(A) 4th period and 10th group (B) 5th period and 10th group
(C) 6th period and 10th group (D) 4th period and 8th group

33. What is the maximum number of electrons in an atom whose highest energy electrons have the principal quantum number n = 5?
(A) 38 (B) 138
(C) 54 (D) None of these

34. A saturated solution of an ionic salt MX exhibits an osmotic pressure of 74.4 mm Hg at 25°C. Assuming that MX is completely dissociated in solution, what is the value of its K_{sp}?
(A) 2×10^{-3} (B) 2×10^{-6}
(C) 4×10^{-6} (D) 4×10^{-3}

35. Which of the following statement is incorrect about the six-coordinated complex $[Fe(ox)_x(H_2O)_y]^z$? (Z may have any value to maintain Fe in +3 oxidation state)
(A) Complex shows optical isomerism, not geometrical isomerism when x = 3, y = 0
(B) Complex shows geometrical isomerism, not optical isomerism when x = 1, y = 4
(C) Complex can show optical and geometrical isomerism both when x = 2, y = 2
(D) Value of z will be – 3 when x = 3, and y = 0

Space for Rough work

36. Human insulin is composed of two polypeptide chains. One chain contains 21 amino acids and the other contains 30 amino acids. How many nucleotides are present in the DNA to code for each chain?
(A) 42, 60 (B) 21, 30
(C) 63, 90 (D) 84, 120

37. Consider the following statements:
(i) All the bond lengths in IF_5 are not equal.
(ii) Equatorial bonds are longer than axial bonds in PCl_3F_2.
(iii) Inorganic benzene is a non polar molecule.
(iv) Order of oxidizing power of oxoacids of chlorine is $HOCl > HClO_2 > HClO_3 > HClO_4$
(v) O – O bond order will be in the order $O_2^+ > O_2 > H_2O_2 > O_3$
(vi) Hydrolysis of XeF_6 is an example of disproportionation reaction.
The correct statements are:
(A) (i) (ii) (iii) (iv) (v) (vi) (B) (i) (ii) (iii) (iv)
(C) (i) (ii) (iii) (iv) (v) (D) (i) (ii) (iii) (iv) (vi)

38. Which of the following is correct order for indicated property?
(A) $PH_3 > PF_3$ bond angle
(B) LiCl > NaCl Solubility in water
(C) F – F > Cl – Cl Bond dissociation enthalpy
(D) S > O > Se Order of $\Delta_{eg}H$

39. Which of the following has the highest boiling point?
(A) H_2 (B) Xe
(C) Ne (D) CH_4

40. Consider the following statements:
(a) Dipole moment order $SO_2 > NH_3$
(b) Ionization energy $O_2 < O$
(c) In naphthalene, all C – C bond lengths are equal
(d) $CF_2 = C = CF_2$ is a non-polar molecule
(e) Cl – CH = C = CH – Cl shows geometrical isomerism
(f) Boiling point $CH_3CH_2OH > CH_3CH_2SH$
(g) Melting point KCl > CuCl
(h) Oxidation number of Cr in $[CrO_8]^{-3}$ is +6.
How many statements are correct?
(A) 5 (B) 4
(C) 7 (D) 6

Space for Rough work

41. A certain reaction $A + B \longrightarrow$ Products is first order with respect to each reactant with K = 5.0 × 10^{-3} M^{-1} s^{-1}. Calculate the conc. of A remaining after 100 s. If the initial concentration of A was 0.1 M and that of B was 6.0 M. (Use approximation of pseudo first order reaction) [$\log_{10}2 = 0.3010$]
(A) 5 × 10^{-3} M (B) 5 × 10^{-2} M
(C) 2.5 × 10^{-4} M (D) 2. 5 × 10^{-2} M

42. Predict the effects upon the following reaction equilibria of increased pressure,
(1) $H_2O(g) \rightleftharpoons H_2(g) + \frac{1}{2}O_2(g)$
(2) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
(3) $C(s) + H_2O(g) \rightleftharpoons H_2(g) + CO(g)$
(4) $\underset{\text{(Diamond)}}{C(s)} \rightleftharpoons \underset{\text{(Graphite)}}{C(s)}$
(A) Left, Left, Left, Left (B) Right, Right, Right, Right
(C) Left, Right, Left, Right (D) Right, Left, Right, Left

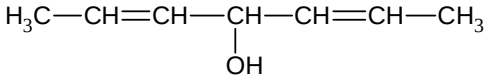
43. Which of the following represent a correct crystal field splitting order for a complex in a linear field. Assume that the ligands lie on the z-axis?
(A) $d_{z^2} < d_{xy} = d_{xz} < d_{x^2-y^2} = d_{yz}$ (B) $d_{xy} = d_{x^2-y^2} < d_{xz} = d_{yz} < d_{z^2}$
(C) $d_{xz} = d_{yz} < d_{z^2} < d_{xy} = d_{x^2-y^2}$ (D) $d_{xy} = d_{x^2-y^2} < d_{xz} = d_{yz} = d_{z^2}$

44. Diborane (B_2H_6) reacts with water to form boric acid and hydrogen. What is the pH of the solution which results when 1.0 g B_2H_6 reacts with 100 mL water? Assume the final volume to be 100 mL. Given K_{a_1} for boric acid = 7.3 × 10^{-10}. (At. Mass of B = 11) [$\log_{10}57 = 1.755$]
(A) 4.64 (B) 5.64
(C) 2.64 (D) 7

45. How many of the following are very good oxidizing agents?
MnO_4^-, I^-, Cl^{-1}, Ce^{+4}, $Cr_2O_7^{-2}$, Na, Na^+, CrO_4^{-2}, HNO_3, Fe^{+2}, F_2, F^-
(A) 5 (B) 6
(C) 7 (D) 4

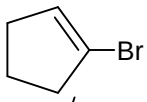
Space for Rough work

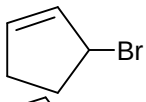
46. Total number of stereoisomers for the following molecule will be:

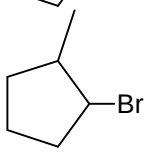


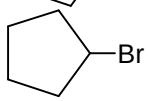
(A) 4
(B) 3
(C) 2
(D) 6

47. Correct order of reactivity of following molecules towards S_N2 mechanism?

(I) 

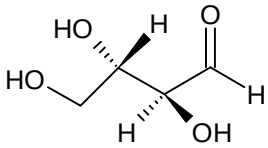
(II) 

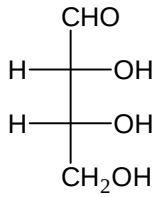
(III) 

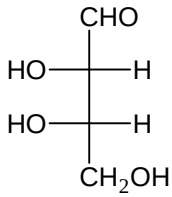
(IV) 

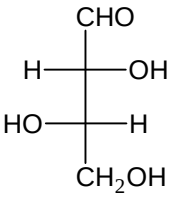
(A) I > II > III > IV
(B) II > IV > III > I
(C) IV > III > II > I
(D) IV > II > III > I

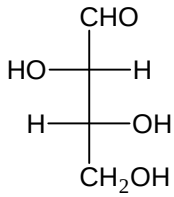
48. Which of the Fischer projection formula correspond to the following stereo structure?



(A) 

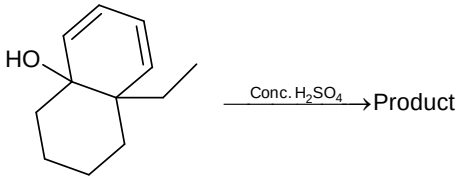
(B) 

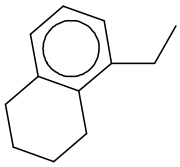
(C) 

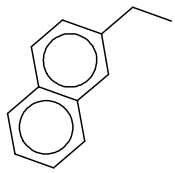
(D) 

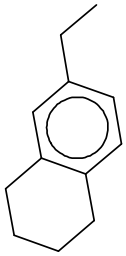
Space for Rough work

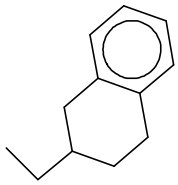
49. Identify the major product



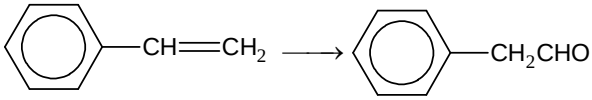
(A) 

(B) 

(C) 

(D) 

50. Which one of the following works best as the reaction steps to carry out conversion below?



(A) Br_2; $NaNH_2$ (2 mole); H_2O, $HgSO_4$ | H_2SO_4 (B) B_2H_6|H_2O_2, NaOH; PCC|CH_2Cl_2
(C) H_2O, H_2SO_4; CrO_3 | H_2SO_4 (D) CH_3COOOH; NaOH | H_2O

51. Which of the following gives silver mirror test?
(I) HCOOH, (II) HCHO, (III) $CH_3 - CH_2 - \underset{\underset{O}{\|}}{C} - CH_2OH$ (IV) $CH_3 - C \equiv CH$, (V) H_3PO_2,
(VI) $CH_2 = CH_2$

(A) I, II, III only (B) I, II, III, IV, V only
(C) I, II, III, V only (D) I, II only

52. A chloro compound (A) showed the following properties:
(i) decolourized Bromine in CCl_4
(ii) absorbed hydrogen catalytically
(iii) gave a precipitate with ammonical cuprous chloride
(iv) when vaporized, 1.49 g of (A) gave 448 mL of vapour at STP.
A is?
(A) $Cl - CH_2 - C \equiv CH$ (B) $CH_3 - C \equiv C - Cl$
(C) $CH_3 - CH = CH - Cl$ (D) $CH_3 - CH_2 - CH_2 - Cl$

Space for Rough work

53. How many of the following can acts as an antiseptic?
(I) C_6H_5COONa (II) 0.1% solution of phenol, (III) Ibuprofen
(IV) potassium stearate, (V) Bithional, (VI) Tincture of iodine
(VII) H_2SO_4, (VIII) $B(OH)_3$, (IX) iodoform, (X) mixture of chloroxylenol and terpeneol
(A) 8 (B) 6
(C) 5 (D) 4

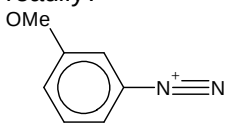
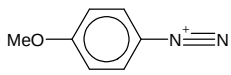
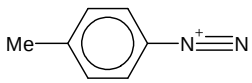
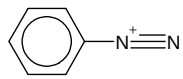
54. Which of the following is an example of condensation homopolymer?
(A) Bakelite (B) Nylon-6,6
(C) Nylon-6 (D) PMMA

55. Diborane shows unsymmetrical cleavage with
(A) Me_3N (B) CO
(C) NH_3 (D) All of these

56. Which of the following statements is incorrect?
(A) The decrease in atomic radii in lanthanide series is not quite regular as it is regular in its M^{+3} ions.
(B) Actinoid contraction is more sharper than lanthanoid contraction.
(C) Ce^{+4} is more stable than Ce^{+3} in aqueous medium.
(D) d-Block elements have greater tendency to show variable oxidation state than f-Block elements.

57. A current of 3.7 A is passed for 6 hours between Ni electrodes in 0.5 L of 2 M solution of $Ni(NO_3)_2$. The molarity of Ni^{+2+} at the end of electrolysis is
(A) 1.172 M (B) 0.172 M
(C) 0.586 M (D) 2 M

58. In $SH^-, I^-, R-NH_2, NH_3$ the order of proton accepting tendency will be
(A) $I^- > NH_3 > R-NH_2 > SH^-$ (B) $NH_3 > R-NH_2 > SH^- > I^-$
(C) $R-NH_2 > NH_3 > SH^- > I^-$ (D) $SH^- > R-NH_2 > NH_3 > I^-$

59. Which of the following diazonium salt when boiled with dil. H_2SO_4 give the corresponding phenol most readily?
(A) 
(B) 
(C) 
(D) 

60. An organic compound A has the molecular formula C_3H_6O. It undergoes iodoform test. When saturated with HCl, it gives B of molecular formula $C_9H_{14}O$. A and B respectively are:
(A) Propanal and mesitylene
(B) Propanone and mesityl oxide
(C) Propanone and 2, 6-dimethyl- 2,5-heptadien-4-one
(D) Propanone and mesityl oxide

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If roots of an equation $y^n - 1 = 0$ are 1, λ_1, λ_2, λ_3 ..... λ_{n-1}, then the value of $(1 - \lambda_1)(1 - \lambda_2)(1 - \lambda_3)$ ..... $(1 - \lambda_{n-1})$ will be
(A) n^n (B) n^2
(C) n (D) 0

62. If $\frac{z_3 - z_1}{z_3 - z_2} = i$, then points z_1, z_2 and z_3
(A) are collinear (B) forms equilateral triangle
(C) forms isosceles right angled triangle (D) forms obtuse angled triangle

63. If α, β and γ are non zero rational numbers, and S is the set of values of $\frac{|\alpha|}{\alpha} + \frac{|\beta|}{\beta} + \frac{|\gamma|}{\gamma}$, then S is
(A) {3, 1} (B) {–3, 1, –1}
(C) {3, –3, 0, 1, –1} (D) {3, –3, 1, –1}

64. If $P = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$, $R = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix}$ and $PQR = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ (where Q is a square matrix of order 2), then |Q| is equal to
(A) $\frac{3}{8}$ (B) $\frac{3}{4}$
(C) $\frac{3}{2}$ (D) 0

65. Let $\cos^2 \alpha + p$ and $\sin^2 \alpha + p$ are the roots of the equation $x^2 + 4x + \frac{61}{16} = 0$. The equation whose roots are $\tan^2 \alpha$ and $\cot^2 \alpha$ is
(A) $x^2 - 4x + 1 = 0$ (B) $x^2 - 10x + 1 = 0$
(C) $x^2 + 10x + 1 = 0$ (D) $x^2 - 14x + 1 = 0$

Space for rough work

66. If n is even number and α, β are the roots of the equation $x^2 + px + q = 0$ and also of the equation $x^{2n} + p^n x^n + q^n = 0$ and $f(x) = \frac{(1+x)^n}{1+x^n}$, then $f\left(\frac{\alpha}{\beta}\right)$ is

(A) –1 (B) 0
(C) 1 (D) 2

67. A 8 ft. tall man finds that the angle of elevation of the top of a 26 ft. high pillar and the angle of depression of its base are complementary angles, the distance of the man from the pillar is

(A) 6 ft. (B) $6\sqrt{3}$ ft.
(C) 12 ft. (D) $12\sqrt{3}$ ft.

68. Number of solution of the $\cos^4 2\alpha + 2\sin^2 2\alpha = 17(\cos\alpha + \sin\alpha)^8$, $\alpha \in (0, 2\pi)$ is

(A) 10 (B) 6
(C) 4 (D) 2

69. If two sides of triangle are roots of $x^2 - 7x + 8 = 0$ and the angle between these sides is 60º then product of inradius and circumradius of triangle is

(A) 8 (B) $\frac{5\sqrt{2}}{3}$
(C) $\frac{5}{3}$ (D) $\frac{8}{7}$

70. Number of ordered pair (α, β), satisfying the condition $\alpha^8 + \beta^8 = 8\alpha\beta - 6$ is

(A) 64 (B) 16
(C) 4 (D) 2

71. If $p_1, p_2, p_3, ..., p_{4001}$ are in A.P. such that $\sum_{i=1}^{4000} \frac{1}{p_i p_{i+1}} = 10$ and $p_1 + p_{4001} = 50$, then $|p_1 - p_{4001}|$ is

(A) 20 (B) 25
(C) 30 (D) 40

72. In a triangle ABC, $R(b + c) = a\sqrt{bc}$ whose R is the circum radius of the triangle, then the triangle is

(A) isosceles but not right angle triangle (B) right angled but not isosceles triangle
(C) right angled isosceles triangle (D) equilateral triangle

Space for rough work

73. If $\vec{v}_1$, $\vec{v}_2$, $\vec{v}_3$, $\vec{v}_4$ are non-zero coplanar vectors and $2\lambda_1\vec{v}_1 + 2\lambda_2\vec{v}_2 - \vec{v}_3 + 3\lambda_3\vec{v}_4 = 0$, then

(A) $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 \geq \frac{1}{2}$ (B) $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 \geq \frac{1}{17}$

(C) $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 \geq \frac{1}{20}$ (D) $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 \geq \frac{1}{34}$

74. Equation of a plane which passes through the point of intersection of lines $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and at greatest distance from the origin is

(A) 7x + 2y + 4z = 54 (B) 4x + 3y + 5z = 50

(C) 3x + 4y + 5z = 49 (D) x + y + z = 12

75. Let $x_1, x_2, x_3, ..., x_n$ be n observations and 5 be its variance, then variance of $2x_1 + 1, 2x_2 + 1, 2x_3 + 1, ..., 2x_n + 1$ is equal to

(A) 20 (B) 6

(C) 5 (D) 25

76. The coefficient of $a^7b^4c^3$ in $(2ab + bc - ca)^7$ is

(A) –560 (B) 560

(C) –280 (D) 280

77. The range of 'k' such that the angles 'α' between the pair of tangents drawn from the point (k, 0) to the circle $x^2 + y^2 = 25$ satisfies $\frac{\pi}{2} < \alpha < \pi$ is

(A) $k \in \left(-\sqrt{2}, -1\right) \cup \left(1, \sqrt{2}\right)$ (B) $k \in \left[-5\sqrt{2}, -5\right] \cup \left[5, 5\sqrt{2}\right]$

(C) $k \in \left(-5\sqrt{2}, -5\right) \cup \left(5, 5\sqrt{2}\right)$ (D) $k \in \left[-\sqrt{2}, -1\right] \cup \left[1, \sqrt{2}\right]$

78. Which of the following statements is a tautology?

(A) $(p \wedge q) \wedge (\sim(p \wedge q))$ (B) $(\sim q \wedge p) \vee (p \vee \sim p)$

(C) $(\sim q \wedge p) \wedge q$ (D) $(\sim q \wedge p) \wedge (p \wedge \sim p)$

79. The number of ways in which 12 identical balls can be grouped in four marked non-empty sets P, Q, R, S such that n(P) < n (Q) (where n(P) is number of balls in group P) is

(A) 70 (B) 60

(C) 45 (D) 25

Space for rough work

80. An ellipse with axes parallel to coordinate axes cuts the parabola $y^2 = 4x$ at (1, –2) and touches it at (4, 4) then the coordinate of other point of intersection is
(A) (4, –4) (B) (3, –2)
(C) (3, 2) (D) (9, –6)

81. An integer a is chosen from {–5, –4, –3, –2, –1, 0, 1, 2, 3, 4, 5}, then the probability that $|a^2 - 3a - 3| > |a^2 + 7a - 13|$ is
(A) $\frac{1}{11}$ (B) $\frac{2}{11}$
(C) $\frac{3}{11}$ (D) $\frac{4}{11}$

82. The number of 3-digit number pqr formed such that $p \neq q \neq r$ and $2q = p + r$ is
(A) 12 (B) 32
(C) 36 (D) 256

83. If a, b, $c \in R$ satisfy the equations $a + [b] + \{c\} = 3{\cdot}1$, $\{a\} + b + [c] = 4{\cdot}3$ and $[a] + \{b\} + c = 5{\cdot}4$ (where [.] denotes the greatest integer function and {.} is fractional part), then $6a + b - 3c$ is
(A) 0 (B) 2
(C) 4 (D) 6.3

84. If $xf(x) + \int_0^x f(t)dt = \frac{3x^2}{2}$ and f(1) = 1, then f(3) is equal to
(A) 1 (B) 3
(C) 9 (D) 28

85. The value of definite integral $\frac{2^{2010}\int_0^1 t^{1004}(1-t)^{1004}dt}{\int_0^1 t^{1004}\left(1-t^{2010}\right)^{1004}}$ is
(A) 4000 (B) 4001
(C) 4020 (D) 4021

86. Consider the function $f(x) = \begin{cases} \frac{g(x)}{x-2} & ; \quad x \neq 2 \\ 7 & ; \quad x = 2 \end{cases}$ where g(x) is a polynomial function such that $g'''(x)$ is identically equal to zero and g(3) = 9. If f(x) is continuous at x = 2, then g(x) is
(A) $2x^2 + x + 6$ (B) $2x^2 - x - 6$
(C) $x^2 + 3$ (D) $x^2 - x + 7$

Space for rough work

87. If $\lim_{t\to\lambda}\left[f(t)+g(t)\right]=2$ and $\lim_{t\to\lambda}\left[f(t)-g(t)\right]=1$, then $\lim_{t\to\lambda} f(t)\cdot g(t)$ is

(A) exist and is $-\frac{3}{4}$ (B) exist and is $\frac{3}{4}$

(C) exist and is $\frac{4}{3}$ (D) does not exist

88. The value of $\lim_{n\to\infty}\sum_{r=0}^{n}\frac{{}^nC_r}{n^r}\left(\frac{1}{r+3}\right)$ is

(A) e (B) e – 1

(C) e – 2 (D) 0

89. If $0 < \cos^{-1} x < \frac{\pi}{2}$ and $1 + \sin(\cos^{-1} x) + \sin^2(\cos^{-1} x) + ... + \infty = 2$, then x is

(A) $\frac{3}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{1}{2}$ (D) $\frac{2}{\sqrt{3}}$

90. The centre of an ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ is C and PN is any ordinate & A, A′ are the end points of major axis, then the value of $\frac{PN^2}{AN\cdot A'N}$ is

(A) 9 (B) $\frac{16}{25}$

(C) $\frac{25}{16}$ (D) 41

Space for rough work